Famicom JAMMA board

power

File: power.kicad_sch

Power: +5V entry, switched +5V_SW, +3V3, supply of the RP2A03 module

cpu

File: cpu.kicad_sch

CPU: RP2A03, work RAM, address decoder, data-bus resistors

ppu

File: ppu.kicad_sch

PPU: RP2C02, address latch, video RAM, sprite detection

cart

File: cart.kicad_sch

Cartridge connector and the dump circuit for emulation mode

video

File: video.kicad_sch

Video: RGB resistor DAC and composite sync

audio

File: audio.kicad_sch

Audio: mixer, loop through the cartridge, power amplifier

jamma

File: jamma.kicad_sch

JAMMA edge connector, inputs, coin counter and lockout drivers

mcu

File: mcu.kicad_sch

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

PWR_FLAG

GND

PWR_FLAG

+5V

PWR_FLAG

+5V_SW

PWR_FLAG

VSYS

Sheet: /
File: fcjamma.kicad_sch

Title: Famicom JAMMA board

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/9

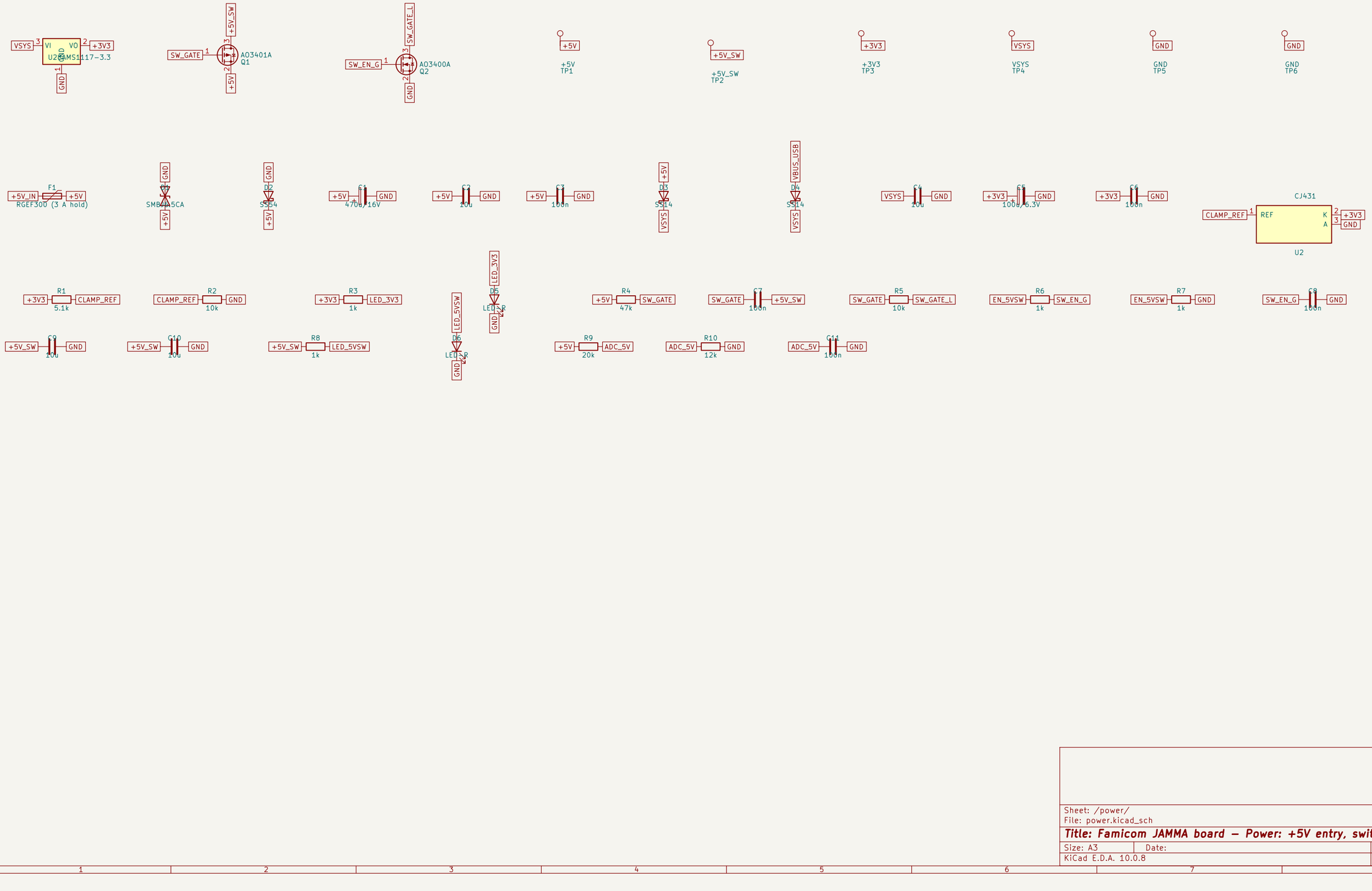
Power: +5V entry, switched +5V_SW, +3V3, supply of the Pico modules

+5V_IN comes from the JAMMA fingers 3, 4, C, D. −5 V and +12 V are not used.

+5V_SW feeds the CPU, the PPU, the memories, the 5 V logic and the cartridge. It is off until Pico C raises EN_5VSW,

so that no 5 V signal reaches a Pico pin before that Pico is powered (the RP2350 pins tolerate 5 V only while powered).

VSYS feeds the three Pico modules and the +3V3 regulator from +5V or from the USB socket, whichever is present.



Sheet: /power/
File: power.kicad_sch

Title: Famicom JAMMA board — Power: +5V entry, switched +5V_SW, +3V3, s

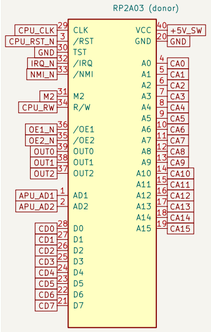
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 2/9

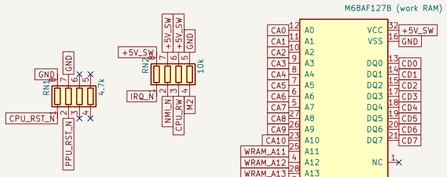
CPU: RP2A03, work RAM, address decoder, data-bus resistors

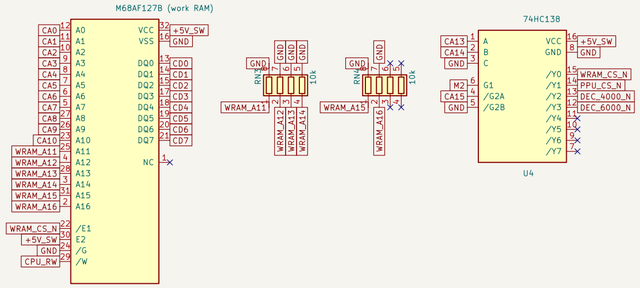
U1 is the donor's RP2A03 in a 40-pin socket. In emulation mode the socket is empty and the dump shift registers drive the buses.

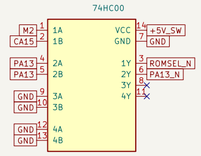
The 470 ohm resistors let Pico B replace what the PPU sees on a write without disturbing the CPU side of the bus.

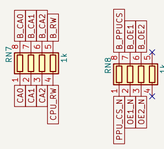
No pull-ups on the CPU data bus: they are known to upset some PPU revisions, and games read the floating bus.

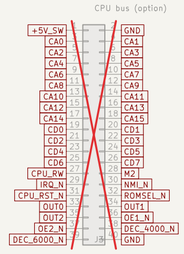






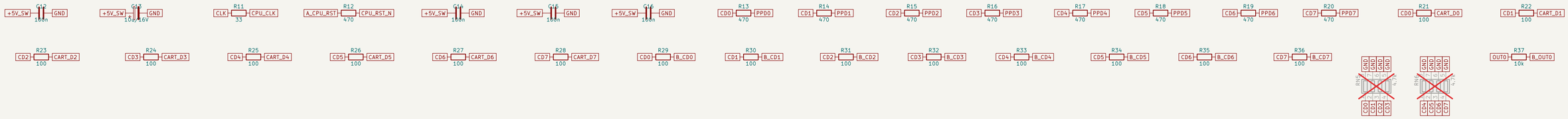


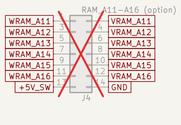












Sheet: /cpu/		
File: cpu.kicad_sch		
Title: Famicom JAMMA board — CPU: RP2A03, work RAM, address decoder, da		
Size: A2	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 3/9

PPU: RP2C02, address latch, video RAM, sprite detector

U6 is the donor's RP2C02 in a 40-pin socket. EXT0–3 give the lower four bits of the palette index of every pixel once Pico B has forced bit 6 of register 2000 high. The fifth bit (sprite or background) is read from the brightness of VOUT: Pico B makes the PPU store a dark colour in every background entry and a bright one in every sprite entry.

RP2C02 (donor)

U6

74HC373

U7

M68AF127B (video RAM)

U8

TL712CP

U9

PPU bus (option)

J4

CMP_OUT

CMP_OUT
TP11

VBUF

VBUF
TP9

VREF

VREF
TP10

MMBT3904
Q3

PPU /RST = +5V (Famicom PPU)

JP1

composite video (test)

J5

sprite bit source

J6

Sheet: /ppu/	
File: ppu.kicad_sch	
Title: Famicom JAMMA board – PPU: RP2C02, address latch, video RAM, sprit	
Size: A2	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 4/9

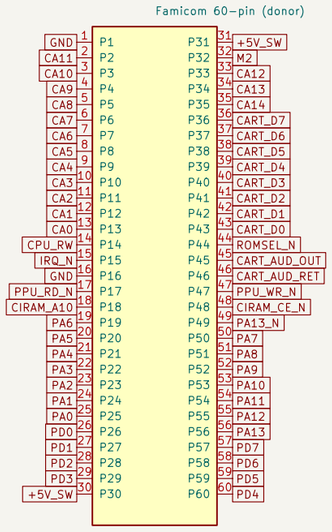
Cartridge connector and the dump circuit for emulation mode

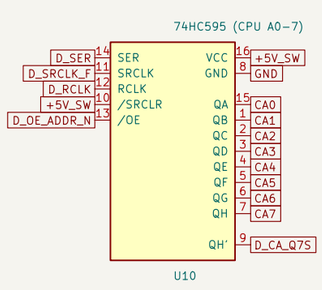
J8 is the 60-pin connector taken from the donor Famicom. Pin 45 carries the board's sound to the cartridge, pin 46 brings it back.

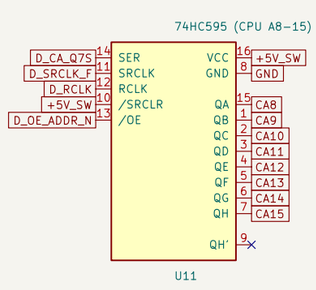
Dump circuit: with the CPU and PPU sockets empty, Pico C reads the cartridge through shift registers and hands the data to Pico A.

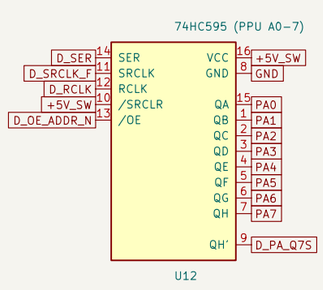
The six 74HC595 run from +5V_SW; Pico C drives their inputs open-drain against 4.7 k pull-ups, so they see proper 5 V levels.

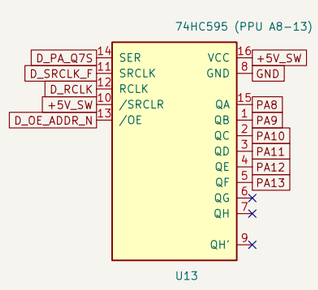
In real-chip mode D_OE_ADDR_N and D_OE_DATA_N stay high (pull-ups) and every 74HC595 output is switched off.

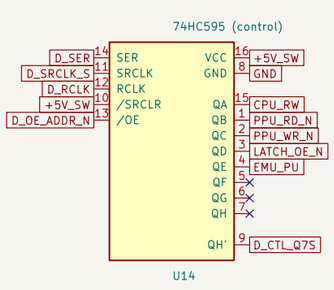


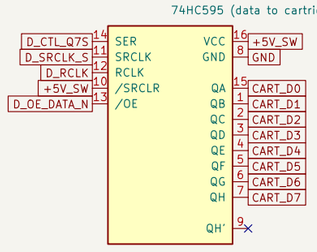


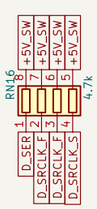


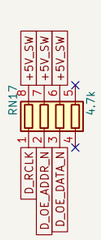


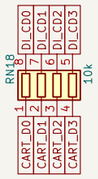








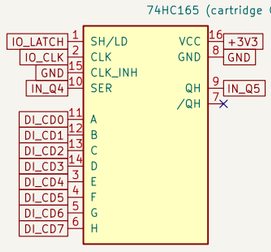


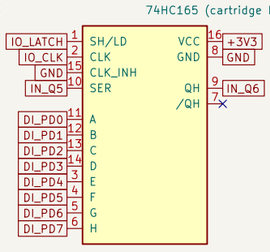


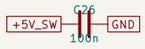


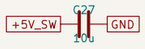


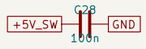


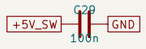


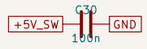


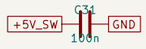




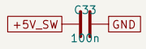




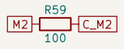


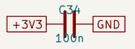


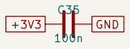


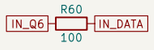


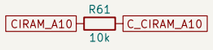














Sheet: /cart/
File: cart.kicad_sch

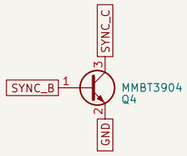
Title: Famicom JAMMA board – Cartridge connector and the dump circuit for

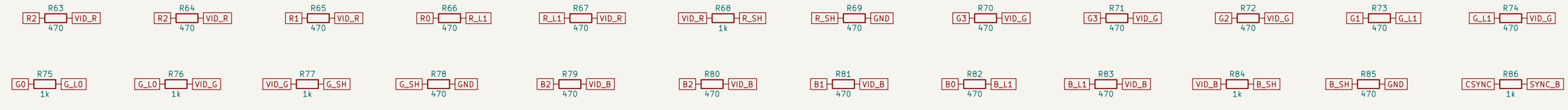
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 5/9

Video: RGB resistor DAC and composite sync

Pico A drives the DAC directly. Full scale is 3.0 V open circuit, about 2.85 V into 2 k, source resistance about 125 ohm.

Red and blue have 3 bits, green has 4. Only 470 ohm and 1 k are used so that the parts list stays short.





RGB + sync (monitor / option)



Sheet: /video/
File: video.kicad_sch

Title: Famicom JAMMA board – Video: RGB resistor DAC and composite sync

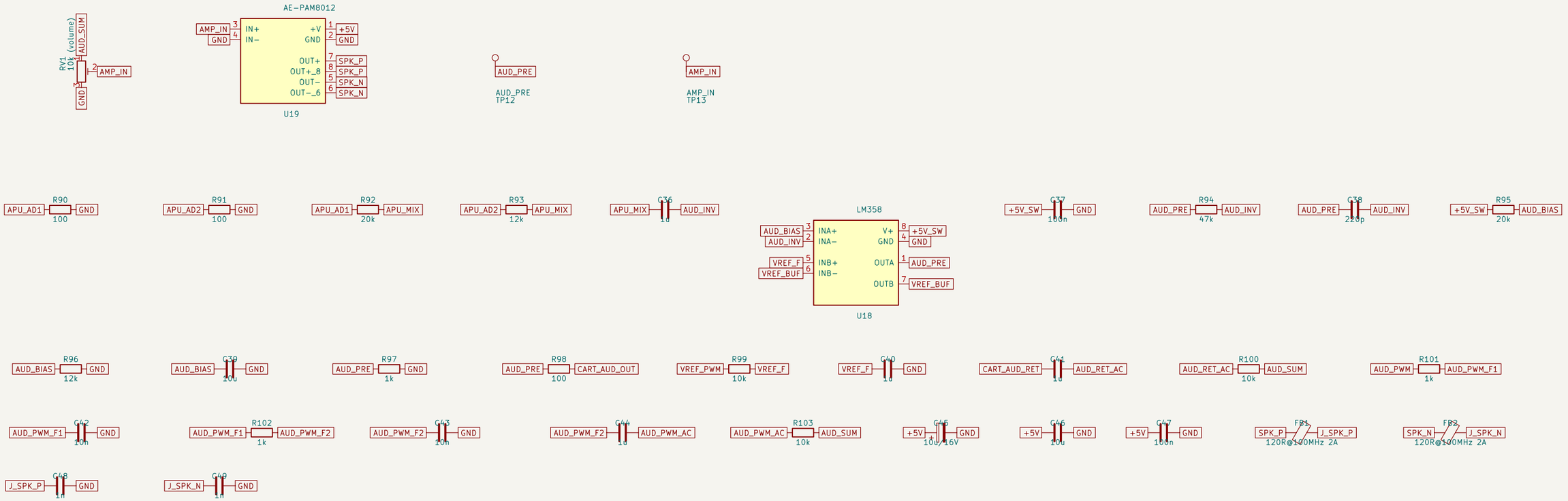
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 6/9

Audio: mixer, loop through the cartridge, power amplifier

Real-chip mode: 2A03 pins -> mixer (Famicom values) -> cartridge pin 45 -> cartridge -> pin 46 -> volume -> amplifier.

Emulation mode: Pico A's PWM output -> two RC stages -> volume -> amplifier.

The loudspeaker outputs are bridge-tied: JAMMA L is not ground.



Sheet: /audio/		
File: audio.kicad_sch		
Title: Famicom JAMMA board – Audio: mixer, loop through the cartridge, pow		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 7/9

JAMMA edge connector, inputs, coin counter and lock-out drivers

Every input: 1 k pull-up to +3V3 (3.3 mA through the switch), 4.7 k in series, 10 nF to ground, then a 74HC165 on +3V3.

Pico C reads the chain: after a low pulse on IO_LATCH the first bit on IN_DATA is input H of U17, the last is input A of U20.

The same pulse copies the 74HC595's shift register to its outputs, so one cycle reads the inputs and writes the outputs.

Fingers 5/E (−5 V), 6/F (+12 V) and 11/M are not connected. 7/H is the key slot.

JAMMA edge (gold fingers)

J1

74HC165 (1P)

U20

74HC165 (2P)

U21

74HC165 (start, coin, service)

U22

74HC165 (switches, status)

U23

74HC595 (JAMMA outputs)

U24

SW1

SW2

SW3

spare outputs (option)

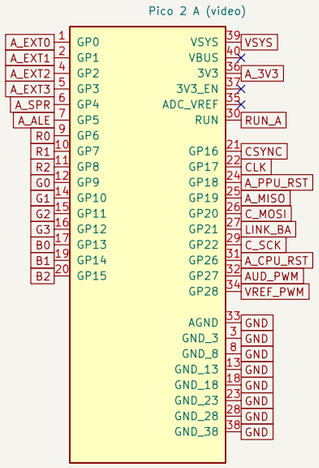
Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

Pico A (video): clock and reset for CPU and PPU, palette index and sprite bit in, RGB and sync out; the emulator in emulation mode.

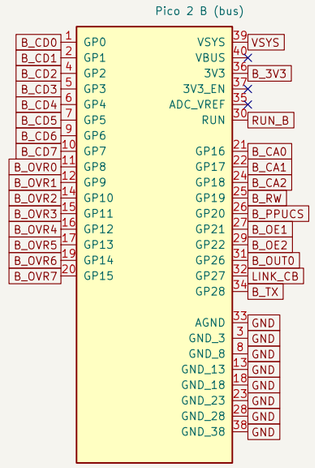
Pico B (bus): watches the CPU's accesses to the PPU, replaces palette data on the PPU side, answers the controller reads.

Pico C (input/output and management): JAMMA inputs and outputs, coin and timer logic, cartridge dump, USB Type-C,

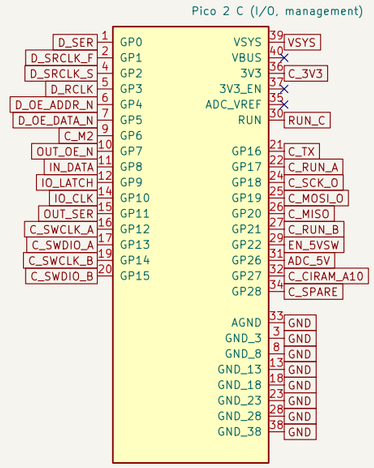
power switch of the Famicom half, and reprogramming of Pico A and Pico B through their debug pins (J13, J14).





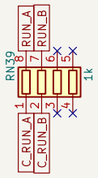


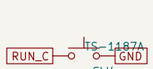




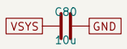


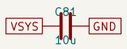


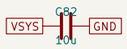


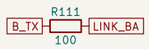




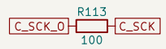


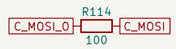


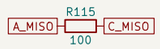




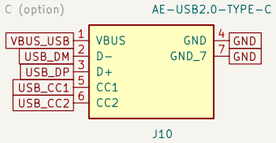


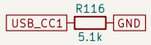


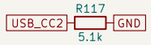


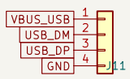




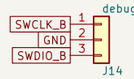


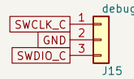












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Title: Famicom JAMMA board – Three Raspberry Pi Pico 2 modules, USB Type

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 9/9